

SENTHILNATHAN KARUPPAIAH

Data Governance & Privacy · Forward-Deployed Engineer · Data & GenAI Architect
linkedin.com/in/senthilsweb · github.com/senthilsweb · senthilsweb.com

nathansweb@icloud.com
732-955-1143
New Jersey, USA

SUMMARY HIGHLIGHTS

- ▶ **Hands-on architect and engineer** who still writes and reviews code daily; **two decades+** building data governance, catalog, discovery, lineage, and privacy platforms for regulated enterprises.
- ▶ **Founding engineer at Io-Tahoe** (data discovery & governance platform, acquired twice) and current **Pentaho Data Catalog** engineering and architecture leadership.
- ▶ Hands-on across **Python, TypeScript, Go, Java, and SQL**, with deep analytics-database work (DuckDB, Snowflake, DBT, lakehouse) — I write and review code daily, not just specs.
- ▶ Privacy & compliance: **PII/NPI detection, GDPR/CCPA/HIPAA, DSR (erase/rectify/send) flows, purpose limitation, retention, access controls, and audit logging** turned into production controls.
- ▶ **Forward-deployed by nature** — I work inside client environments translating messy real-world data and regulatory requirements into working solutions, with the customer-facing judgment to land them.
- ▶ GenAI & agentic engineering: **spec-driven AI-DLC methodology, multi-agent orchestration (eve, Pydantic AI), Presidio-based LLM privacy, and a model mix spanning Claude, GPT-4, GLM, DeepSeek, and local Ollama.**
- ▶ **Automation-oriented** — much of my work is the kind of process automation once called RPA, now rebuilt with GenAI and deterministic tool-calling.
- ▶ **Open-source and community contributor** — publishing utilities, demo apps, and regular LinkedIn articles on data governance, privacy, and GenAI; blogging since 2007.
- ▶ **Global delivery experience** across the US, Japan (3 yrs, conversational Japanese), India, and Ukraine; led distributed teams across Asia, Europe, and the Americas.

CORE COMPETENCIES

Data Governance & Privacy	Discovery, Catalog & GenAI	Platform & Engineering
Privacy by Design · Data Minimization	PII/NPI Detection · Classification	Python · TypeScript · Go · Java
GDPR · CCPA · HIPAA · DSR Workflows	Data Catalog · Metadata · Lineage	DuckDB · Snowflake · DBT · SQL
Purpose Limitation · Retention · Audit	Agentic Analytics · Semantic Layer	AWS/S3 · Kubernetes · CI/CD
LLM Anonymization · Presidio	AI-DLC · eve · Pydantic AI · RAG	BentoML · vLLM · APIs · Observability

PROFESSIONAL EXPERIENCE

Senior Engineering Manager & Data Governance / Privacy Architect | Pentaho

Remote, NJ May 2021 – Present

- ▶ Lead hands-on engineering for Pentaho Data Catalog (PDC) and PDI/BA — discovery, metadata, classification, lineage, access-policy, and governed analytics across enterprise customer environments.
- ▶ Architected automated S3/lakehouse cataloging and metadata harvesting via PDC REST APIs; delivered BRDs, specs, swimlanes, and client-ready controls for regulated financial services customers.
- ▶ Built privacy/sensitive-data discovery for PII/NPI at scale: profiling, data-subject search, classification rules, human review, audit trails, and DSR erase/rectify/send operating flows.
- ▶ Designed NLP/ML metadata enrichment and classification routing (Legal-BERT via BentoML) plus GenAI gateway patterns turning privacy requirements into production discovery services.
- ▶ Pioneered the AI-DLC methodology and Pentaho SDET AI Workbench — a spec-driven agentic engineering practice with deterministic tool use and SQL traceability, adopted as a team standard.
- ▶ Partnered with CDO, CISO, privacy, customer success, and professional-services stakeholders on governance, retention, access control, anomaly detection, and observability requirements.

Solutions Architect / Client Delivery & Head of Production Engineering | Io-Tahoe LLC (Consultant)

New York City, NY Dec 2018 – May 2021

- ▶ Founding team member of Io-Tahoe, an early AI-driven data discovery and governance product company acquired twice; shaped catalog/discovery from product ideation through enterprise delivery.
- ▶ Delivered data governance, catalog, lineage, metadata, and compliance programs across banking, insurance, and healthcare; converted regulatory requirements into implementable discovery workflows.
- ▶ Led client delivery and production engineering for sensitive-data discovery, data-subject request mapping, access/audit controls, reliability, and release readiness across the product suite.

Solutions Architect | AZ-Teach Radiology & Open MRI

Mesa, AZ Jul 2017 – Nov 2018

- ▶ Architected custom Radiology Information System (RIS), HL7 integration engine, and web/mobile apps for multi-site radiology centers.

EARLIER EXPERIENCE

Io-Tahoe — Solution Architect

New York, NY | Dec 2014 – Jun 2017

Platform engineering and product ideation for the Io-Tahoe data discovery product (Java/J2EE, Python).

Vmoksha Technologies — Solutions Architect

NJ, USA & Bangalore, India | Aug 2008 – Jun 2017

Delivered enterprise applications (incl. Dole Fresh Vegetables); led teams across US and India.

Digitalpia — Programmer Analyst

Matsuyama, Japan | Jan 2006 – Aug 2008

Multiple web and client-server projects using Oracle, MS SQL Server, and MS Access.

Infosys Technologies — Team Leader

Bangalore, India | Jun 2004 – Dec 2005

Solution engineering, technology consultation, and performance optimization for enterprise clients.

Hewlett Packard — Systems Engineer

Bangalore, India | Oct 2003 – Jun 2004

Workforce Management Product and SMART CRAWLER for Information Security.

Earlier Roles (1999 – 2003)

Bangalore, India

Software engineering across call-center, CMS, e-commerce, and telecom directory systems.

INDEPENDENT PROJECTS & OPEN-SOURCE AT github.com/senthilswb.

templrgo — Portable Data & Ops Toolkit

- ▶ Single-binary, config-driven developer platform with Lua serverless functions, SQL playgrounds, embedded BoltDB storage, and an AI chat assistant; powers senthilswb.com (github.com/senthilswb/templrgo).

ai-agents — Vercel Eve Agent Monorepo

- ▶ Monorepo of small Vercel Eve agents sharing one runtime kit; filesystem-first design where typed tools do the deterministic work, models only coordinate, and every run records data, report, and metrics to a timestamped folder.

OpenLineage Proxy Server

- ▶ Next.js-based OpenLineage proxy that receives and stores lineage events as individual JSON files for debugging and analysis.

Privacy Shield — LLM Privacy Engineering Utility

- ▶ Open-source Python/FastAPI utility using Presidio, LangChain, and OpenAI/Ollama to detect and anonymize PII before LLM processing, restore it afterward, and preserve full auditability.

Adaptive Data Governance IP & LinkedIn Publishing

- ▶ Publishes long-form LinkedIn articles, cheatsheets, and carousels on data governance and GenAI under #AdaptiveDataGovOps; blog at senthilswb.com (since 2007);

AI-Assisted Clinical Trial Data Platform *(Life Sciences / HealthTech — Independent Build)*

- ▶ Independently architected a governed clinical data platform with agentic AI development — eCRF design, trial management, automated pipelines, compliance
- ▶ Built agentic analytics: AI Chat over trial data, governed semantic layer, audit trails, DBT pipelines, EHR/CTMS/ePRO integration — Next.js, DuckDB, DBT, LangChain, MinIO.

AI & GENAI TOOLING

Models & Platforms: Claude, GPT-4, Gemini, GLM, DeepSeek, Ollama/local models

Privacy/AI Tooling: Presidio, LangChain, PII detection, reversible de-identification

Frameworks: eve, Pydantic AI, Flue, pgvector, Knowledge Graphs, RAG

Serving: BentoML, vLLM, HuggingFace, Cloudflare Workers AI

Methodology: Spec-first, AI-DLC, privacy-first local inference, deterministic tools

Data AI: Catalog, lineage, classification, semantic layers, DBT, Cube.dev, text-to-SQL

EDUCATION & CERTIFICATIONS

B.E. (Electronics & Instrumentation Engineering) & M.B.A (IT & Tourism)

Annamalai University, India

Social Innovation Leader – Global (SIL-G)

Hitachi Academy · IMD Business School

Microsoft Certified Professional

Web Applications, Database Design, Windows Dev — MS .NET & SQL Server

Global Experience

Japan (3 yrs, conversational Japanese) · India · Ukraine